Distributed GLR: factorization, spectral operations, and memory layouts

Design record for the GLR engine of `lgpsf-hessian` (the `lgh_glr_*` API; ScaLAPACK backend).
Originally written 2026-08-16 as the design record of the engine's first production deployment, a large-scale ice-sheet inversion; the outer factors Z, M are the library's prior interface ($R = ZM^{-1}Z$, `lgh_prior_t`).
Supersedes the replicated-T structure documented in `glr-woodbury-formulas.pdf` (which the library keeps as its replicated backend).

1 Operators and factorization

N dofs, distributed by the application's spatially-local mesh partition over P ranks. M = diagonal lumped mass. Z = the prior's square-root factor (e.g. a shifted Laplacian at a reference regularization scale); Z^{-1} available as a linear operator (multigrid-preconditioned Chebyshev, `lgh_prior_create_mat`). Prior precision $H_r = ZM^{-1}Z$ (bilaplacian-like). $\widetilde{H}_d$ = symmetrized lgpsf approximation of the misfit Hessian (stage 1, `lgh_fit_*`), available through matvecs on mesh-partitioned vectors. c = regularization / Levenberg–Marquardt weight.

$$\widetilde{H}(c) = \widetilde{H}_d + c\,H_r = ZM^{-1/2}\,(F + cI)\,M^{-1/2}Z, \tag{1}$$

$$F = M^{1/2}Z^{-1}\widetilde{H}_d\,Z^{-1}M^{1/2} = F^T \qquad \text{(whitened prior-preconditioned misfit Hessian; never formed).} \tag{2}$$

Keep c and the regularization scale α explicit; do not absorb them into Z. Reasons: (i) one eigendecomposition of F serves every c through the spectral filters below (the library's operations all take c per call), whereas c absorbed into Z would force a rebuild per shift; (ii) for the bilaplacian with $Z = \alpha A$, changing α rescales eigenvalues exactly with unchanged eigenvectors, $F(\alpha') = (\alpha/\alpha')^2 F(\alpha)$ — a free update, visible only if α is a scalar in the code.

2 Target operations

1. Remove spurious negative eigenvalues of F (approximation artifacts): FLIP treatment $\lambda_i \leftarrow |\lambda_i|$ (`LGH_TREAT_FLIP`, the default). Do not discard large-magnitude negatives — they are genuine flipped-curvature modes, and clipping them to zero leaves $\mathcal{O}(1 + |\lambda|)$ outliers in the preconditioned spectrum.

2. Solves with the (positively modified) $\widetilde{H}$: the deterministic MAP phase (`lgh_glr_solve`).

3. Square-root and inverse-square-root *factor* actions for MCMC sampling (`lgh_glr_factor`, `lgh_glr_sample`).

4. Log-determinant differences $\log\det \widetilde{H}_1 - \log\det \widetilde{H}_2$ for detailed balance with location-dependent proposal precision (`lgh_glr_logdet`).

3 Randomized eigendecomposition of F

Sketch width ℓ (target rank $r \le \ell$ after truncation; continental-scale expectation in the originating application $r \sim 10^4$–2×10^4, must scale to 5×10^4+).

$$\Omega_{ij} = \mathrm{hashnormal}(\mathrm{seed}, i, j) \quad \text{(drawn locally from global index; partition-independent)} \tag{3}$$

$$Y \leftarrow F\,\Omega; \qquad \text{for } p = 1, \dots, q: \quad Q \leftarrow \mathrm{orth}(Y),\; Y \leftarrow FQ \tag{4}$$

$$Q \leftarrow \mathrm{orth}(Y) \qquad \text{(panel-blocked BCGS2; no } \ell \times \ell \text{ Gram is ever materialized)} \tag{5}$$

$$T = \mathrm{sym}(Q^T F Q) \qquad \text{(formed in column panels, reduced directly into 2D block-cyclic layout)} \tag{6}$$

$$T = V \Lambda_{\mathrm{raw}} V^T \qquad \text{(distributed dense eigensolver: ScaLAPACK } \texttt{pdsyevd}\text{, ELPA later; } V \textit{ stays in place}\text{, block-cyclic; } \Lambda \text{ replicated)} \tag{7}$$

$$U = QV_{:,\mathcal{K}} \quad \textit{implicit}\text{: } Uw = Q(Vw),\;\; U^T x = V^T(Q^T x); \qquad \lambda_i = |\lambda_{\mathrm{raw},i}| > \tau \;\; \text{(FLIP + cut)} \tag{8}$$

$U^T U = I_r$; $F \approx U \Lambda U^T$. The coefficient traffic per U-apply is $\mathcal{O}(\ell)$ numbers — negligible. At large ℓ the implicit form is forced, not merely elegant: materializing $U = QV$ with block-cyclic V would require either replicating V ($8\ell^2$ bytes per rank; 20 GB at $\ell = 5 \times 10^4$) or a heavyweight mixed-layout distributed GEMM.

Incremental rank growth (first-class operation: `lgh_glr_extend`*).* Since r is not known a priori: extend by a block of k_{new} fresh hashed columns (global column index continues past the current ℓ, so the extension is deterministic and partition-independent), apply F, BCGS2-orthogonalize against the existing Q, border T with $Q^T(FQ_{\mathrm{new}})$, re-eigensolve. Stop when the sketch's smallest $|\lambda|$ (reported as `next_abs`) falls below the truncation cut τ with margin — the unknown rank becomes a stopping criterion instead of a gamble on ℓ. (Evidence this matters: in the originating application, a Pine Island Glacier sketch at $\ell = 2200$ was rank-limited — 2136 of 2200 modes survived the cut.)

4 Spectral operations

With the truncated tail of F treated as 0 (the GLR model itself), any spectral function acts as

$$f(F + cI) \;=\; U[f(\Lambda + cI) - f(c)\, I_r]U^T \;+\; f(c)\, I \qquad (\texttt{lgh\_glr\_filter}). \tag{9}$$

Only the actions of U and U^T are required.

Solve (MAP phase). $f(x) = 1/x$ in (**??**) gives

$$\widetilde{H}(c)^{-1} = Z^{-1} M^{1/2} \tfrac{1}{c}\big(I - U D_c U^T\big)\, M^{1/2} Z^{-1}, \qquad D_c = \mathrm{diag}\left(\frac{\lambda_i}{\lambda_i + c}\right), \tag{10}$$

(`lgh_glr_solve`), which at $c = 1$ is exactly the Woodbury apply of the engine's originating production code (continuity check). The inverse is the one case where the whitened spectral function composes exactly with the outer factors, because it is the inverse of a product.

Square-root factors (MCMC). The sandwich is *not* a similarity transform, so the symmetric square root $\widetilde{H}^{1/2}$ is not cheaply available — and is not needed. Sampling requires a factor:

$$G = ZM^{-1/2}(F + cI)^{1/2}, \qquad \widetilde{H}(c) = G\,G^T, \tag{11}$$

with $(F + cI)^{\pm 1/2}$ from (**??**). All four factor actions ($G,\ G^T,\ G^{-1},\ G^{-T}$; `lgh_glr_factor`) reduce to U/U^T applies, $M^{\pm 1/2}$ scalings, and Z matvecs or solves. A draw from the proposal $\mathcal{N}(m, \widetilde{H}^{-1})$ is $x = m + G^{-T}\xi,\ \ G^{-T} = Z^{-1}M^{1/2}(F + cI)^{-1/2},\ \ \xi \sim \mathcal{N}(0, I)$ (`lgh_glr_sample`).

Log-determinant differences. Z and M are state-independent and cancel between two builds:

$$\log\det \widetilde{H}_1(c) - \log\det \widetilde{H}_2(c) = \sum_k \log\bigl(\lambda_k^{(1)} + c\bigr) - \sum_k \log\bigl(\lambda_k^{(2)} + c\bigr) \tag{12}$$

where the per-build quantity to compute (`lgh_glr_logdet`) is $\sum_k[\log(\lambda_k + c) - \log c]$ over the *kept, treated* eigenvalues: truncated modes contribute 0, so the sum is finite and truncation-consistent, and the common Nc-dependent constant cancels in the difference. *Detailed-balance consistency:* Metropolis–Hastings needs the log-determinant of the proposal precision *actually sampled from.* Sampling and log-determinant both read the same treated Λ, so truncation and FLIP do not threaten correctness — they only affect proposal quality (acceptance rate).

5 Shapes and memory layouts

Feature-space principle: the reduced space $\mathbb{R}^\ell$ is distinct from the function space $\mathbb{R}^N_M$ and its dual, and owns its own partition — by feature index (2D block-cyclic), never by mesh geometry. Small feature-space objects (Λ, filter diagonals) are replicated; that is a storage convention, not a leak between spaces.

Symbol	Shape	Space	Layout
v, x, w_{full}	N	function / dual	mesh partition (N_p per rank)
$M^{\pm 1/2}$	N (diag)	—	mesh partition
Z	$N \times N$ sparse	dual $\leftarrow$ function	mesh partition; Z^{-1} via MG-Chebyshev
$\widetilde{H}_d$	$N \times N$ sparse	dual $\leftarrow$ function	mesh partition (MPIAIJ)
Ω, Y, Q	$N \times \ell$ dense	whitened	mesh-partitioned row slabs ($N_p \times \ell$)
BCGS2 coefficients	$\ell \times b$	—	transient, replicated per panel (discarded)
T	$\ell \times \ell$	feature	2D block-cyclic on $p_r \times p_c$ grid; never replicated
V	$\ell \times \ell$	feature	block-cyclic, *stays in place* after eig
Λ, D_c, filters	r (diag)	feature	replicated (small)
$U = QV_{:,\mathcal{K}}$	$N \times r$	whitened $\leftarrow$ feature	*implicit composition; never formed*
w (reduced vectors)	ℓ or r	feature	conformal with T's layout; gathered ($\mathcal{O}(\ell)$ traffic) inside U-applies
c, α, τ	scalars	—	replicated

Memory at the ambition point ($N = 10^7,\ \ell = 5 \times 10^4,\ P = 100$ nodes): T, V are 20 GB *aggregate* (∼0.2 GB/node, block-cyclic) — no longer a constraint. The tall blocks dominate: Q is 4 TB aggregate (40 GB/node), and the sweep buffers must be managed (overwrite in place, panel the F sweeps). At $N = 10^6$ everything is comfortable.

6 Design rationale (brief)

- **All three $\ell \times \ell$ objects go distributed, not just T.** The orthogonalization Gram matrix has the same $\mathcal{O}(\ell^2)$ replicated footprint as T; panel-blocked BCGS2 avoids materializing it at all, is more robust than CholQR for power-iterated (ill-conditioned) blocks, and makes incremental column-block extension natural.

- T **is formed directly into its block-cyclic home**, in column panels (transient $\ell \times b$ per rank), never assembled replicated.

- **Implicit** U avoids both the $N\ell r$ materialization GEMM and any replication of V; per-apply cost is essentially unchanged (GEMV over ℓ columns of Q vs. r columns of U, plus $\mathcal{O}(\ell)$ coefficient traffic).

- **Incremental growth** turns unknown rank into a stopping criterion; supporting $r \gtrsim 5 \times 10^4$ cheaply is a core advantage of lgpsf-based approximation over conventional low-rank methods and must not be blocked by a fixed-ℓ pipeline.

- **Eigensolver**: ScaLAPACK `pdsyevd` first (ships with MKL on large clusters; reference build available everywhere via PETSc), ELPA/SLATE behind the same seam when $\ell \gtrsim 3 \times 10^4$ makes it pay. Eigenvector signs / near-degenerate rotations depend on the process grid, so cross-P (and cross-backend) comparisons use invariants only (Λ, operator applies, logdet).

- **Determinism**: hashed Ω (and hashed extension columns) keep the sketch partition-independent; a dedicated study of T on the originating problem showed it is a generic dense symmetric matrix (no banding, no hierarchical structure, full-rank off-diagonal blocks), so a dense distributed eigensolver is the right primitive — there is nothing to exploit by compression.